

Part 5: Coupled Oscillators

Baseline:

Mass m (kg)	0.05
Single-spring k (N/m)	26.92
$\omega_0^2 = k/m$ (rad ² /s ²)	538.4432487
$\omega_1^2 = (3+\sqrt{5})/2 \cdot \omega_0^2$	1409.662726
$\omega_2^2 = (3-\sqrt{5})/2 \cdot \omega_0^2$	205.66702
$f_1 = \omega_1/2\pi$ (Hz)	5.975548007
$f_2 = \omega_2/2\pi$ (Hz)	2.282456237

Equal masses – Mode 1	Average Measured f (Hz)	$\omega = 2\pi f$	ω^2	Theoretical ω^2	% diff
	2.2143	13.91285723	193.5675962	205.66702	5.88

Equal masses – Mode 2	Average Measured f (Hz)	$\omega = 2\pi f$	ω^2	Theoretical ω^2	% diff
	5.8333	36.65170485	1343.347469	1409.662726	4.70

Unequal masses – Heavy on top, Mode 1	Average Measured f (Hz)	$\omega = 2\pi f$
	1.9286	12.11775118

Unequal masses – Heavy on top, Mode 2	Average Measured f (Hz)	$\omega = 2\pi f$
	4.6905	29.47128068

Unequal masses – Light on top, Mode 1	Average Measured f (Hz)	$\omega = 2\pi f$
	1.7143	10.77126457

Unequal masses – Light on top, Mode 2	Average Measured f (Hz)	$\omega = 2\pi f$
	5.4286	34.10889976

Same result when reversed? **Yes (within 20%)**